

Behavioral Results

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_all_losses

Tcl Console Messages Log

Using 2 slave threads.
Starting static elaboration
Pass Through NonSizing Optimizer
Completed static elaboration
INFO: [XSIM 43-4323] No Change in HDL. Linking previously generated obj files to create kernel
INFO: [USF-XSim-69] 'elaborate' step finished in '2' seconds
INFO: [USF-XSim-4] XSim::Simulate design
INFO: [USF-XSim-61] Executing 'SIMULATE' step in 'C:/posit_loss_project/vivado/vivado_build/posit_loss_project.sim/sim_1/behav/xsim'
INFO: [USF-XSim-98] *** Running xsim
with args "tb_all_losses_behav -key {Behavioral:sim_1:Functional:tb_all_losses} -tclbatch {tb_all_losses.tcl} -view {C:/posit_loss_project/vivado/vivado_build/tb_all_lo
INFO: [USF-XSim-8] Loading simulator feature
Time resolution is 1 ps
open_wave_config C:/posit_loss_project/vivado/vivado_build/tb_all_losses_time_synth.wcfg
source tb_all_losses.tcl
set curr_wave [current_wave_config]
if { [string length \$curr_wave] == 0 } {
if { [llength [get_objects]] > 0 } {
add_wave /
set_property needs_save false [current_wave_config]
} else {
send_msg_id AddWave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->New Waveform
}
}
run 1000ns
==== FULL SYSTEM TEST ====
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_all_losses_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:06 ; elapsed = 00:00:08 . Memory (MB): peak = 1669.469 ; gain = 22.156
run all
MSE Loss: 0x18 | MSE Grad: 0x30
CE Loss: 0x36 | CE Grad: 0xd0
\$finish called at time : 10725 ns : File "C:/posit_loss_project/verilog/tb_all_losses.v" Line 48

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_all_losses

Scope Sources Objects Protocol Instanc

tb_all_losses_time_synth.wcfg tb_all_losses.v

Name	Value	Data Type
clk	0	Logic
rst	0	Logic
start_mse	0	Logic
start_ce	0	Logic
y_hat[7:0]	48	Array
y_true[7:0]	40	Array
p_hat[7:0]	30	Array
label	1	Logic
mse_g[7:0]	18	Array
mse_g[7:0]	30	Array
ce_l[7:0]	36	Array
ce_g[7:0]	d0	Array

tb_all_losses_time_synth.wcfg

Name	Value
clk	0
rst	0
start_mse	0
start_ce	0
y_hat[7:0]	48
y_true[7:0]	40
p_hat[7:0]	30
label	1

Tcl Console Messages Log

run 1000ns
==== FULL SYSTEM TEST ====
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_all_losses_behav' loaded.
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Scope x Sources

Name	Design Unit	Block Type
tb_all_losses	tb_all_losses	Verilog M

Objects x Protocol Instance

Name	Value	Data Type
clk	0	Logic

tb_all_losses.v x tb_all_losses_time_synth.wcfg

C:/posit_loss_project/verilog/tb_all_losses.v

Tcl Console x Messages Log

```
INFO: [USF-XSim-96] *** Running xsim
with args "tb_all_losses_behav -key {Behavioral:sim_1:Functional:tb_all_losses} -tclbatch {tb_all_losses.tcl} -view {C:/posit_loss_project/vivado/vivado_build/tb_all_losses.tcl}
INFO: [USF-XSim-8] Loading simulator feature
Time resolution is 1 ps
open_wave_config C:/posit_loss_project/vivado/vivado_build/tb_all_losses_time_synth.wcfg
source tb_all_losses.tcl
# set curr_wave [current_wave_config]
# if { [string length $curr_wave] == 0 } {
#   if { [llength [get_objects]] > 0 } {
#     add_wave /
#     set_property needs_save false [current_wave_config]
#   } else {
#     send_msg_id Add_Wave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->New Waveform'"
#   }
# }
# run 1000ns
==== FULL SYSTEM TEST ====
INFO: [USF-XSim-96] XSim completed. Design snapshot 'tb_all_losses_behav' loaded.
INFO: [USF-XSim-97] XSim simulation ran for 1000ns
launch_simulation: Time (s): cpu = 00:00:03 ; elapsed = 00:00:06 . Memory (MB): peak = 4651.355 ; gain = 0.000
run all
MSE Loss: 0x18 | MSE Grad: 0x30
CE Loss: 0x36 | CE Grad: 0xd0
$finish called at time : 10725 ns : File "C:/posit_loss_project/verilog/tb_all_losses.v" Line 48
```

Type a Tcl command here

Sim Time: 10725 ns

Finished Writing Synthesis Report : Time (s): cpu = 00:00:40 ; elapsed = 00:00:43 . Memory (MB): peak = 1962.871 ; gain = 1183.070

Synthesis finished with 0 errors, 0 critical warnings and 0 warnings.

Synthesis Optimization Runtime: Time (s): cpu = 00:00:30 ; elapsed = 00:00:41 . Memory (MB): peak = 1962.871 ; gain = 1115.715

Synthesis Optimization Complete : Time (s): cpu = 00:00:40 ; elapsed = 00:00:43 . Memory (MB): peak = 1962.871 ; gain = 1183.070

INFO: [Project 1-571] Translating synthesized netlist

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.018 . Memory (MB): peak = 1972.023 ; gain = 0.000

INFO: [Netlist 29-17] Analyzing 98 Unisim elements for replacement

INFO: [Netlist 29-28] Unisim Transformation completed in 0 CPU seconds

INFO: [Project 1-570] Preparing netlist for logic optimization

INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).

Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.001 . Memory (MB): peak = 1976.676 ; gain = 0.000

INFO: [Project 1-111] Unisim Transformation Summary:

No Unisim elements were transformed.

Synth Design complete | Checksum: f0ca5c31

INFO: [Common 17-83] Releasing license: Synthesis

48 Infos, 2 Warnings, 0 Critical Warnings and 0 Errors encountered.

synth_design completed successfully

synth_design: Time (s): cpu = 00:00:45 ; elapsed = 00:00:50 . Memory (MB): peak = 1976.676 ; gain = 1440.512

Write ShapeDB Complete: Time (s): cpu = 00:00:00 ; elapsed = 00:00:00.006 . Memory (MB): peak = 1976.676 ; gain = 0.000

INFO: [Common 17-1381] The checkpoint

'C:/posit_loss_project/vivado/vivado_build/posit_loss_project.runs/synth_1/top_wrapper.dcp' has been generated.

INFO: [Vivado 12-24828] Executing command : report_utilization -file

top_wrapper_utilization_synth.rpt -pb top_wrapper_utilization_synth.pb

INFO: [Common 17-206] Exiting Vivado at Mon Apr 20 19:34:18 2026...